(Sisqual) Results Sisqual Problems

October 13, 2026

1 Experimental Setup

- **CPU:** 11th Gen Intel® Core™ i7-1165G7 @ 2.80 GHz × 8
- **Memory and OS:** [15Gi], [22.04.5 LTS (Jammy Jellyfish)]
- **Software:** [Python 3.10.12], PuLP [3.3.0], CBC [2.10.3]

2 Optimizers

Solvers (ILP) : CBC

-> Reference solution : The Exact ILP was solved with a gap limit of 1% and 4 Threads.

3 Restrictions Covered

- Full coverage of minimum staffing requirements per time slot and competence (*soft constraint*);
- At most one schedule and one competence assignment per employee per day (*Hard Constraint*);
- At least 11 hours of rest between consecutive schedules (*Hard Constraint*);
- Each employee must be assigned only to the competences they are allowed to perform (*Hard Constraint*);
- Employee unavailability (vacation, sick leave, forced day off) must not be violated (*Hard Constraint*);
- Fixed-time schedules (`EQUALS:`) must be respected (*Hard Constraint*).

4 Cases Description

The case being compared is **Sisqual_October_2025** and **Sisqual_Complete**.

5 Problem Specification and Models

5.1 Minimum / Ideal / Estimated Staffing

The problem under study consists of generating work schedules for a set of employees while satisfying operational requirements and legal constraints. The main objective is to ensure adequate staffing targets for each competence and time interval throughout the planning horizon. Specifically, three staffing targets are considered: minimum, ideal, and estimated staffing levels.

For each competence and time interval, the schedule should assign a number of employees that meets, whenever possible, the corresponding minimum, ideal, and estimated staffing requirements. Any violation of these requirements is penalized in the objective function. The overall objective is to minimize the number of unmet staffing requirements according to a hierarchical priority structure.

5.2 Competences and Levels

All employees are assigned specific competences that they perform during their schedules, and these competences can be switched multiple times throughout the day. Every competence has a proficiency level associated with it per employee, meaning that every individual possesses a specific degree of proficiency for each of their assigned competences (where, for instance, level 1 denotes maximum proficiency). This structure introduces an extended optimization objective that prioritizes a hierarchical order of proficiency levels that must be successfully accomplished alongside operational coverage.

5.3 Constraints

Besides staffing coverage, the model includes several structural constraints, described in Section 3. These constraints ensure that the generated schedules comply with both operational requirements and labor regulations.

5.4 Models

One methodology was developed to address the problem: a approach based entirely on pure heuristics. The greedy heuristic is described in the companion document *(Sisqual) Greedy Heuristic for Multi-Competence / Multi-Level Scheduling*; in this report it is referred to as the *Greedy Heuristic With Levels*.

In summary, the problem combines staffing requirements, legal constraints, and individual employee availability limitations, resulting in a highly complex combinatorial optimization problem whose solution requires appropriate techniques to balance solution quality and computational efficiency.

5.5 Metrics

The following metrics are reported (lower is better in all cases):

- **Total Shortage:** total number of unmet minimum staffing requirements, summed over all days, time slots and competences;
- **Staffing Ideals:** the same measure computed against the ideal staffing levels;
- **Staffing Estimated:** the same measure computed against the estimated staffing levels;
- **Slots With Failure:** total number of slots that miss an employee by 1 or many employees (this value is always less or equal than the total shortage value).

This greedy heuristic targets exclusively the minimum staffing requirements; the remaining metrics are reported for reference.

6 Results

Base results are obtained with the optimal configuration for the problems described in Section 4.

Sisqual_October_2025

The base algorithm produced the following results in **0.68 seconds**:

Metric	Value
Total Shortage	511
Staffing Ideals	493
Staffing Estimated	493
Slots With Failure	489

Table 1: Base results for Sisqual_October_2025.

Results Summary - Hybrid Heuristic With Levels

Metric	Value
Total Shortage	579
Staffing Ideals	555
Staffing Estimated	555
Slots With Failure	561

Table 2: Hybrid Heuristic With Levels: best result for Sisqual_October_2025.

Time(s)	Total Shortage	Staffing Ideals	Staffing Estimated	Slots With Failure
0.183	615.4	591.4	591.4	597.4

Table 3: Hybrid Heuristic With Levels: average results for Sisqual_October_2025.

Variance presents a value of **89.12** for all metrics evaluated

Note on the measurement basis. The average values were computed over 16000 attempts of the heuristic. *Time(s)* is the average time of one attempt.

Sisqual_Complete

The base algorithm produced the following results in **1.1 seconds**:

Metric	Value
Total Shortage	120
Staffing Ideals	120
Staffing Estimated	120
Slots With Failure	120

Table 4: Base results for Sisqual_Complete.

Results Summary - Hybrid Heuristic With Levels

Metric	Value
Total Shortage	542
Staffing Ideals	542
Staffing Estimated	542
Slots With Failure	542

Table 5: Hybrid Heuristic With Levels: best result for Sisqual_Complete.

Time(s)	Total Shortage	Staffing Ideals	Staffing Estimated	Slots With Failure
0.2	574.8	574.8	574.8	574.8

Table 6: Hybrid Heuristic With Levels: average results for Sisqual_Complete.

Variance presents a value of **79.68** for all metrics evaluated

Note on the measurement basis. The average values were computed over 16000 attempts of the heuristic. *Time(s)* is the average time of one attempt.

7 Comparative Analysis

This section compares the reference solution (exact ILP, 1% gap) with the *Greedy Heuristic With Levels* on the two instances. Since the heuristic targets exclusively the minimum staffing requirements, the comparison is driven by the *Total Shortage* metric (lower is better); the other metrics follow the same behaviour.

7.1 Solution Quality

Table 7 reports the relative gap of the best and of the average heuristic solution to the reference.

Instance	Method	Total Shortage	Gap	Time (s)
Sisqual_October_2025	ILP (reference)	511	0.0%	0.68
	Heuristic (best)	579	13.3%	–
	Heuristic (average)	615.4	20.4%	0.183
Sisqual_Complete	ILP (reference)	120	0.0%	1.1
	Heuristic (best)	542	351.7%	–
	Heuristic (average)	574.8	379.0%	0.2

Table 7: Relative gap to the reference solution on the Total Shortage metric. Time is per attempt for the heuristic.

Three observations follow.

- **The heuristic is reasonably close on October problem, but not on Complete.** On Sisqual_October_2025 the best solution is 13.3% above the reference, whereas on Sisqual_Complete it is about 4.5 times the reference value (542 against 120). The reference solution covers almost all the minimum demand of Complete, which indicates that a very good solution exists but that a local, one-employee-at-a-time construction does not find it. A plausible explanation is that each block is chosen without looking at how it affects the employees processed afterwards, so shortages that a global model would avoid by redistributing blocks are locked in early; this hypothesis was not tested.
- **The gap is structural, not a matter of luck.** The variance of 89.12 (October) and 79.68 (Complete) corresponds to a standard deviation of about 9.4 and 8.9 shortages, i.e. roughly 1.5% of the mean. The distance between the average and the best attempt (36.4 and 32.8) is much smaller than the distance between the best attempt and the reference (68 and 422).
- **Randomisation has a limited effect.** Random tie-breaking and the random shuffle inside each rarity group provide diversity, but all attempts follow the same greedy logic and end up in a narrow band of values.

7.2 Computational Efficiency

A single heuristic attempt (0.183 s and 0.2 s) is only about 3.7 and 5.5 times faster than the reference ILP (0.68 s and 1.1 s), and on both instances the ILP is already solved in about one second.

7.3 Scalability

- **Exact ILP.** It is fast on these instances, but the number of variables grows with employees, days, time slots and competences, and the multi-level structure may make it much harder for larger workforces or longer horizons; this was not tested here.
- **Greedy heuristic.** Its cost is polynomial in the number of employees, days and candidate blocks and does not depend on a solver. Its advantage would appear for instances where the ILP becomes too slow or too memory-demanding, at the cost of a solution quality that, on these instances, is considerably worse.

8 Conclusions

On the two instances tested, the exact ILP is clearly the best method: it is fast (0.68 s and 1.1 s) and provides much lower shortages, so the heuristic is not justified by performance here. The heuristic achieves a moderate gap on Sisqual_October_2025 (13.3% for the best solution, 20.4% on average) but a large one on Sisqual_Complete (351.7% and 379.0%), and since the variation between attempts is small, the gap is a limitation of the construction itself and not of the randomness. Its value lies in larger or more constrained instances, where the exact model may become impractical, and as a fast way to obtain an initial feasible schedule.

9 Limitations and Future Work

- Results come from **two instances**; conclusions about scalability should be validated with more and larger instances (more employees, competences and days).
- The heuristic is **purely constructive**, and decisions are never revised. Adding a **local search / repair phase** is the most natural way to reduce the gap.
- A **hybrid approach**, in the spirit of the Puzzle Heuristic used in the 3-shift problem (small ILPs over windows of days combined with greedy phases), or using the heuristic solution as a **warm start** for the ILP, may combine speed and quality.
- The **ideal and estimated** staffing levels are not used by the construction; incorporating them as secondary objectives would make the heuristic cover the full hierarchy.
- Analyse **where the remaining shortage concentrates** (by competence, time slot and day) and the sensitivity to the **day-order mode**, in order to understand the large gap on Sisqual_Complete.